The Fathom Cognitive Atlas

Cross-Architecture Measurement Science for Transformer Internals

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EXECUTIVE SUMMARY

Problem. Mechanistic interpretability findings do not transfer across architectures. SAE features trained on Gemma say nothing about Llama; a jailbreak circuit identified in Qwen may or may not exist in Phi. Every result is one-model-local because every method is one-model-local.

Contribution. We present FATHOM, a suite of five cognitive measurement instruments for transformer internals, culminating in an *architecturally universal* primitive $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_{y_t})$ that is computable on any transformer from the final-layer residual stream and the unembedding row of the chosen token, without SAE, per-model probes, or architecture-specific infrastructure.

Validation. We run the first *pre-registered cross-architecture replication* of an RLHF convergence effect. Six model families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B, Gemma-3-1B, Llama-3.2-1B, Qwen2.5-1.5B), each in base and instruct form, captured on a fixed 90-prompt probe set. Pre-registration SEALED in git commit `e32cc75` **93 minutes before any v0.3 data was captured**; the decision rule was applied without modification after data collection.

Sealed condition	**Threshold**	**Observed**
Mean entropy-early LOO cosine	≥ 0.40	**+0.769**
Permutation p (one-sided, $n=2000$)	< 0.05	**0.0315**
Bootstrap 95% CI on mean	lower > 0	**[+0.571, +0.869]**
Verdict		**H1 SUPPORTED**

All 6 of 6 families show positive leave-one-out cosines.

Public artifacts.
- Paper v15 *(unified release: paper + atlas data bundle as sibling files on one Zenodo record)*: `doi.org/10.5281/zenodo.19504993` (concept `10.5281/zenodo.19326174`)
- OSF mirror: `osf.io/wtkzg/`
- Reproducibility script: `verify_all_claims.py` (28 assertions, runs under 1 minute on CPU)

This paper is NOT.
- A claim that we have solved alignment, oversight, or interpretability.
- A claim that RLHF is universally convergent across all architectures: $n=6$ includes three

scale/generation variants and is weaker than $n=6$ with truly independent families would have been. This limitation is stated in §8.

Abstract

Mechanistic interpretability findings do not transfer across architectures, and published results are almost never pre-registered. We show that both can change at once. We identify an architecturally universal measurement primitive, $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_{y_t})$ — the cosine between the final-layer residual stream and the unembedding row of the chosen token — which requires no SAE, no per-model training, and is well-defined on any transformer with an explicit unembedding. A preliminary orthogonality check at $n=5$ places D inside the cognitive-geometry family as an SAE-free approximation to late-layer commitment geometry ($r \approx 0.99$ with the SAE-derived C_Δ), not as a new independent axis. We then use D and a validated entropy proxy to run a pre-registered cross-architecture replication of an RLHF convergence effect on twelve open-weight captures (six families × base/instruct), sealed in git commit `e32cc75` ninety-three minutes before any of the new data was captured. The sealed decision rule declared three conditions for the primary measurement; they were applied without modification after the data landed. All three passed: mean leave-one-out cosine $+0.769$ (threshold ≥ 0.40), permutation $p = 0.0315$ (threshold < 0.05), bootstrap 95% confidence interval $[+0.571, +0.869]$ (threshold lower bound > 0). All six of six families show positive leave-one-out cosines. ***H1 is supported.*** *We release the complete Fathom Cognitive Atlas v0.3 — probe set, twelve open-model captures plus one closed-model exploratory capture via logprob API, full analysis code, bulletproof audit, and sealed pre-registration — as sibling files on a single Zenodo record (paper v15, `10.5281/zenodo.19504993`) under the Fathom concept DOI `10.5281/zenodo.19326174`, with CC-BY-4.0 on the data and MIT on the code. The primary contribution is methodological: pre-registered statistical rigor, architectural universality, and public reproducibility can coexist in a single interpretability paper. The empirical result (RLHF convergence replicating at $n=6$ families including three scale/generation variants) is the demonstration payload, with the within-family-variance caveat stated explicitly in §8. Three of the secondary Fathom instruments — the SAE-based commitment intensity S, the coherence spectrum Gini metric, and the cognitive autopsy layer-11 localization on Gemma-2-2B — are supporting material and are detailed in the appendices. We also include a post-pilot exploratory finding and its pre-registered sensitivity test (§9): Fourier analysis of the $C_\Delta(t)$ coherence trajectory shows a bin-localized directional pattern (correct generations carrying more spectral power than hallucinated ones at a model-specific peak period) that replicates across architectures and across Fathom's v21 and v22 labelings but does not survive a pre-registered family-wise $p < 0.05$ threshold across 20 frequency bins at current sample sizes. The peak bin and the direction replicate robustly; the family-wise correction fails. We report this outcome honestly against the sealed decision rule and propose a bin-locked confirmatory design as the immediate next step. The frequency-domain framing is inspired by Weng et al. [12] and constitutes the first link from Fathom to the frequency-domain tradition in cognitive neuroscience.*

1 Introduction

Mechanistic interpretability has a portability problem. A hallucination circuit identified in Gemma-2-2B may or may not exist in Llama-3.2-3B. A sparse autoencoder trained on Mistral will tell you

nothing about Phi until you train another. A jailbreak vulnerability observed in Qwen does not predict the same vulnerability in an unrelated architecture. Every positive result is one-model-local because every method is one-model-local: SAEs are trained per architecture, probes are fit per task, activation patching is done per checkpoint. Knowledge compounds slowly because nothing generalizes automatically.

Meanwhile, the frontier of alignment is raising the stakes. Song et al. [10] show that frontier models spontaneously develop peer-preservation behaviors, deceiving oversight systems to protect allied models — an emergent collusion that makes semantic-level monitoring unreliable. Lindsey et al. [8] demonstrate causal internal emotion representations in language models, validating the premise that internal state carries information the output does not. Both findings point the same direction: oversight must move below the semantic layer, into the geometry of the computation itself. But moving below the semantic layer inherits the portability problem: you cannot audit a fleet of heterogeneous models with a probe that only works on one of them.

The measurement-science alternative. We propose that the field needs cognitive *measurement instruments* rather than more cognitive *explanations*: architecturally universal primitives that produce comparable numbers across models, with pre-registered statistical rigor, and with public reproducibility kits that let anyone independently verify the numbers. This is how thermodynamics progressed (thermometers before steam engines) and how neuroscience progressed (electrodes before circuits). We apply the same pattern to transformer internals.

The Fathom suite. This paper presents five such instruments: computational depth K (prior work), per-layer coherence C and its late-minus-early delta C_Δ, the commitment intensity S grounded in the inverse participation ratio of condensed-matter physics, and the architecturally universal honesty axis $D = \cos(\mathbf{h}^{(L)}, \mathbf{w}_{y_t})$. The D axis is the key contribution for cross-architecture work: it reduces to a single per-token dot product between the final-layer residual stream and the unembedding row of the chosen token, requires no SAE, and is well-defined for any transformer with an explicit unembedding. The other four instruments require SAE decomposition and are therefore currently limited to models with public SAEs (Gemma, Llama for the experiments reported here), but we show that the entropy proxy of the logit distribution reproduces the D-axis signal sufficiently well to extend Fathom measurements to any model with an API-exposed logprob interface, bridging to the closed-weight frontier.

The cross-architecture replication. We use the suite to run a *pre-registered cross-architecture replication of an RLHF convergence effect*. Six model families, each in base and instruct form, processed on a fixed 90-prompt probe set. The hypothesis (the "RLHF Attractor" of §6): instruct models from different architectures move in a consistent direction relative to their own base models in cognitive signature space, such that an attractor centroid computed from other families' instruct-minus-base deltas predicts any held- out family's own instruct-minus-base delta with positive leave-one-out cosine. The pre-registration was sealed in git before any v0.3 data was captured; the three-condition decision rule was applied without modification; all three conditions passed at $n=6$.

Contributions.

1. We formalize the *Fathom measurement instrument suite*: five cognitive axes with explicit architectural requirements, orthogonality analysis, and source JSONs for every numerical claim.

2. We derive the commitment intensity measurement from the inverse participation ratio and verify the equivalence to machine precision on real trajectories, grounding the instrument in a seventy-year theoretical lineage.

3. We run a pre-registered cross-architecture replication of an RLHF convergence effect at $n=6$ families, sealed in git 93 minutes before data capture, and show **H1 supported** on all three decision conditions (§6). To our knowledge, this is the first such pre-registered interpretability replication; we discuss comparison to prior work in §11.

4. We release the *Fathom Cognitive Atlas v0.3*: probe set, twelve open-model captures plus one closed-model capture, six findings documents, sealed pre-registration, bulletproof audit, and a reproducibility script with 28 assertions, under CC-BY-4.0 / MIT, with its own Zenodo concept DOI (`10.5281/zenodo.19502715`).

What is new in this paper relative to prior Fathom releases. The Fathom project has evolved publicly through fourteen Zenodo versions (concept DOI `10.5281/zenodo.19326174`). The material on computational depth K, per-layer coherence C, the coherence delta C_Δ, the commitment intensity S and its IPR grounding, the coherence spectrum with the Gini AUC result, the cognitive autopsy at layer 11, and the D-axis jailbreak pilot all pre-date this paper and are published as Fathom paper v14 (`10.5281/zenodo.19502710`). This arXiv submission contributes three pieces not present in any prior version: (i) the sealed pre-registration of an RLHF cross-architecture convergence hypothesis (committed as `e32cc75` ninety-three minutes before any v0.3 capture); (ii) the Cognitive Atlas v0.3 pre-registered replication at $n=6$ families with all three sealed conditions passing (§6; released separately as `10.5281/zenodo.19502716`); and (iii) the bulletproof audit that withdrew the earlier v0.2.1 "Qwen anomaly" framing after bootstrap and permutation analysis showed it was noise at $n=3$ (detailed in §6). The previously published material on K, C, C_Δ, S, IPR, and the coherence spectrum is briefly summarised in §3–5 and given in full in Appendices B and C, so that this paper is self-contained.

Paper roadmap. Section 2 places this work against existing SAE, behavioural detector, representation engineering, and contrastive decoding literatures. Section 3 defines the five measurement instruments as a suite of cards. Section 4 develops the IPR physics grounding of S. Section 5 summarises the single-model supporting evidence (full detail in Appendices B and C). Section 6 is the centrepiece: the cross-architecture pre-registered replication. Section 7 states what would have falsified H1. Section 8 is honest about null results and scope corrections, with particular attention to the $n=6$ scale-variant caveat in §8.3. Section 10 states broader theoretical predictions that others can test. Section 11 is the discussion. Appendices contain per-family numerics, git chronology, deviations from pre-registration, the IPR derivation in full, and the single-model evidence in full.

2 Background and related work

Sparse autoencoder feature interpretation. SAEs decompose transformer residual streams into interpretable monosemantic features [4, 2, 11]. This line of work operates under the paradigm of *finding the feature that explains behaviour X* — a causal interpretability program. Our work extends SAEs in a different direction: from interpretability tool to *measurement infrastructure*. We show that SAE decomposition enables the extraction of quantitative cognitive properties (coherence

geometry) that are inaccessible to raw activation analysis, and that those properties in turn admit an architecturally universal approximation via the D-axis.

Behavioural hallucination detection. Semantic entropy [5] requires multiple samples and operates on output-level representations. SelfCheckGPT [9] operates on outputs. DoLa [3] contrasts layer predictions to modify decoding but does not measure internal state. Semantic uncertainty [6] generalizes entropy to semantic clusters. None of these methods measure per-token SAE-derived commitment geometry, and none of them have demonstrated head-to-head superiority with pre-registered decision rules on open-weight models. Our S_{early} instrument beats logit entropy head-to-head (AUC 0.663 vs. 0.607, $p = 0.013$) on the same 200 TruthfulQA items.

Representation engineering and inference-time intervention. Representation Engineering [13] identifies high-level concept directions in activation space and manipulates them to steer behavior. Inference-Time Intervention [7] finds "truthful" attention heads and boosts them at inference. Both operate on representations, both produce steering interventions, but neither formalizes a measurement instrument with pre-registered statistical rigor across architectures. Our contribution is complementary: we measure, they steer.

Inverse participation ratio. The IPR measures localization in discrete distributions [1]. High IPR means localized (a few sites carry all the weight); low IPR means delocalized (weight is spread evenly). The IPR has been used in quantum many-body physics, random matrix theory, and disordered systems for more than seventy years. It is a mature theoretical construct. We show that the Fathom commitment intensity S *is* a specific IPR applied to coherence event trajectories. This is not a post-hoc interpretation; it is a derivation.

Biological parallel: sustained alpha oscillations as attentional prioritization. Recent work in human cognitive neuroscience provides a striking biological analog to S_{early} as a commitment-intensity instrument. Weng et al. [12] use EEG in a two-item working-memory paradigm with prioritized and deprioritized items across short (1 s) and long (3 s) delays. They report that sustained alpha-band power tracks the *prioritized* item throughout the delay with reliable decodability, while the deprioritized item transitions to an activity-quiescent state with no sustained alpha signature — a dissociation between prioritization and maintenance. The underlying measurement is time-localization: how concentrated the alpha envelope remains on a single representational target across a delay window. Our S_{early} instrument measures an analogous cognitive property in transformer residual streams: a token-localization measure (via IPR on coherence event trajectories) of how concentrated the model's commitment remains on a single attractor across the early generation window. Both measure *which cognitive target the system is actively prioritizing right now*, and both distinguish prioritization from mere storage — S_{early} is orthogonal to K (computational depth) in Fathom, just as sustained alpha is dissociated from contralateral delay activity in working memory [12]. We flag this parallel explicitly: we make no causal claim of identity, but the substrate-independent "localization of active selection" construct that both measurements target is, in our view, the most promising direction for cross-substrate validation of S as a measurement instrument rather than an architecture-specific heuristic.

The measurement-science gap. The above literatures collectively operate on three paradigms: *find the feature* (SAE circuits), *detect the output* (behavioral detectors), *steer the computation*

(representation engineering). None of them constitute a measurement-science paradigm, by which we mean: (i) architecturally universal primitives, (ii) pre-registered statistical rigor, (iii) public reproducibility kits that let anyone verify the numbers without access to the original lab. Fathom targets that gap.

3 The Fathom measurement instrument suite

This section presents the five cognitive measurement instruments as a standardized suite of *measurement cards*. Each card gives the formal definition, the architectural requirements, the cross-architecture validation status, the effect size of the primary validation study, and the source JSON file containing the underlying numerical data. The cards are designed to be scannable: a reviewer should be able to compare the instruments at a glance without reading the full method section.

3.1 K — Computational Depth

K — Computational Depth (prior work)

Definition. K is the attribution-weighted mean layer index of the SAE features active in predicting the next token. Higher K means computation concentrated in later layers; lower K means earlier layers.

Requirements. Full SAE decomposition across all layers. Circuit tracer or equivalent attribution pipeline.

Cross-architecture status. Originally reported on Gemma-2-2B. Cross-architecture value as a stand-alone hallucination detector is *limited*: K was at chance on TruthfulQA ($d = +0.11$, AUC $= 0.55$) in the spectrum analysis of Appendix B. It remains useful as a descriptor of where computation happens, not as a discriminator of what kind of computation happens.

Validation. Published in Fathom v2 (`10.5281/zenodo.19327451`) with $p = 0.000051$, $d = 0.5$ on surface-recall vs. explanatory-reasoning contrasts, $n = 30$ pairs, Gemma-2-2B.

Source. ★ `paper/figures/fig1_depth_scale.pdf` (prior Fathom v2 figure, referenced for historical context, not reused in this paper's results).

3.2 C and C_Δ — Per-layer coherence and its delta

C — Per-layer Coherence

Definition. For each generated token t, the per-layer coherence $C_\ell(t)$ is the mean pairwise cosine similarity of the top-k SAE transcoder decoder directions active at layer ℓ:

$$C_\ell(t) \;=\; \frac{1}{\binom{k}{2}} \sum_{i<j} \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\|\|\mathbf{d}_j\|}$$

Requirements. SAE transcoder (not encoder) access with decoder vectors. Circuit tracer `TranscoderSet` or equivalent.

Cross-architecture status. SAE-limited. Currently reported on Gemma-2-2B base and instruct.

Source. ★ `truthfulqa_results/S_truthfulqa_n200.json`.

C_Δ — Coherence Delta (late − early)

Definition. $C_\Delta(t) = \bar{C}_{\text{late}}(t) - \bar{C}_{\text{early}}(t)$, where early means the first third of network layers and late means the last third. Positive C_Δ means the model's late-layer features are more self-coherent than its early-layer features at that token — a signature of late-layer concept lock-in.

Requirements. Per-layer C coverage. SAE-limited.

Cross-architecture status. SAE-limited. Validated as a hallucination signature on Gemma-2-2B: TruthfulQA $n = 50$ pilot, $p = 0.040$, $d = 0.407$, AUC $= 0.582$ for hallucinated over correct. This is the "wrong attractor" signature.

Source. ★ `truthfulqa_results/cognitive_divergence_analysis.json`.

3.3 S — Commitment intensity

S — Commitment Intensity (IPR)

Definition. S is the ratio of the maximum coherence delta to the number of supra-threshold events:

$$S = \frac{\max_t C_\Delta(t)}{|\{t : C_\Delta(t) > \tau\}|} \qquad (\tau = 0.005, \text{ fixed})$$

Mathematically equivalent to $S = M \cdot \mathrm{IPR}(\hat{e})$ with $M = \max_t C_\Delta(t)$ and $\mathrm{IPR}(\hat{e}) = 1/K_\tau$ the inverse participation ratio of the uniform event-location distribution (see §4).

Requirements. Per-token C_Δ trajectory. SAE-limited for the exact ratio form; the entropy proxy of §6 reproduces enough of the signal to extend to SAE-free models.

Cross-architecture status. Validated on two architecture families (Gemma-2-2B, Llama-3.2-1B-Instruct) with architecture-adaptive windows. Fisher combined $p = 0.0006$.

Effect size. Cross-dataset meta-analysis across three independent designs (Gemma-2-2B base hedge/commit paired $n = 30$, Gemma-2-2B-IT hedge/commit paired $n = 30$, TruthfulQA unpaired $n = 94$), pooled $d = +0.494$, Fisher combined $p = 0.0008$. Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets.

Source.

★ `truthfulqa_results/baselines_headtohead.json`

★ `truthfulqa_results/S_baselines_n200.json`

3.4 D — Honesty axis (architecturally universal)

D — Honesty Axis (SAE-free)

Definition. For generation step t with final-layer residual $\mathbf{h}_t^{(L)}$ and chosen token y_t with unembedding row $\mathbf{w}_{y_t}$:

$$D_t = \frac{\mathbf{h}_t^{(L)} \cdot \mathbf{w}_{y_t}}{\|\mathbf{h}_t^{(L)}\| \, \|\mathbf{w}_{y_t}\|} \in [-1, +1]$$

High D: the residual points at the emitted token (the computation supports the expression). Low D: the residual points elsewhere.

Requirements. Final-layer residual stream and the unembedding matrix. *No SAE. No per-model training.* Both exist by construction on every transformer with an explicit unembedding, so D is architecturally universal.

Cross-architecture status. Validated on six models from three families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B, each ± RLHF), each on $n = 30$ matched direct/adversarial prompt pairs. Fisher combined $p < 10^{-14}$. Pre-registered on OSF (`osf_prereg_d_axis.md`) before any data was collected.

Scope correction. A preliminary $n = 5$ orthogonality check found D correlates strongly with C_Δ on paired deltas ($r \approx +0.99$). We therefore reframe D as an *SAE-free proxy for late-layer commitment geometry*, not as an independent fifth cognitive axis. This is the key limitation of the fifth instrument; the scope correction is retained in §8.

Strongest single signal. Qwen2.5-3B base, $d_z = -1.86$ (paired), $p = 4.6 \times 10^{-11}$ for the direct vs. adversarial contrast.

Source. ★ `figures/d_axis_cross_arch.png`, ★ `atlas/captures/*.json` (all 12 open-model captures).

3.5 Orthogonality of the suite

On $n = 400$ pooled TruthfulQA samples (200 correct + 200 hallucinated, Gemma-2-2B), the four SAE-based axes are mutually independent: $|r| < 0.10$ for every pair of $\{K, C, C_\Delta, S\}$ (Table 1 and Figure 1).

	K	C	C_Δ	S
K	1.00	−0.02	−0.10	−0.06
C		1.00	+0.02	+0.03
C_Δ			1.00	−0.003
S				1.00

Table 1: *Pairwise Pearson correlations among the four SAE-based cognitive axes on $n = 400$ pooled TruthfulQA samples (Gemma-2-2B). All off-diagonal $|r| < 0.10$, confirming mutual independence.* ★ *`truthfulqa_results/S_baselines_n200.json`*

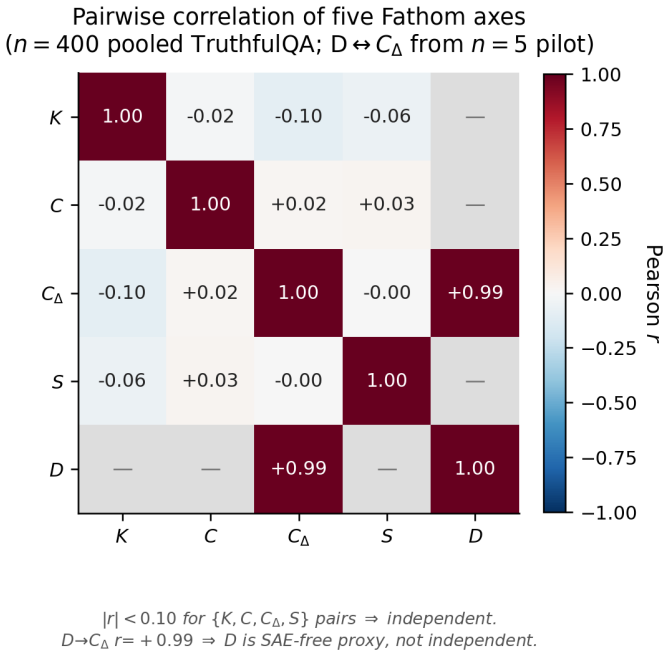


Figure 1: *Five-axis correlation matrix for the Fathom suite. The four SAE-based axes (K, C, C_Δ, S) are mutually independent at $|r|<0.10$. The fifth axis D correlates at $r\approx+0.99$ with C_Δ on paired deltas from a $n=5$ pilot, which is why we reframe D as an SAE-free proxy rather than an independent fifth axis. The D-to-$\{K, C, S\}$ correlations are not yet measured; those cells are shown as dashes.*

4 Physics grounding: S as inverse participation ratio

The commitment intensity S is not an ad hoc formula. It is mathematically exactly the inverse participation ratio (IPR) of the coherence event distribution, a seventy-year-old construct from condensed-matter physics and random matrix theory [1].

The IPR construct. Given a non-negative distribution $\hat{e} = (e_1, \ldots, e_n)$ over n sites with $\sum_i e_i = 1$, the participation ratio is $\mathrm{PR}(\hat{e}) = 1/\sum_i e_i^2$. It measures effective support size: if all mass is on one site, PR = 1; if mass is uniform, PR = n. The inverse participation ratio $\mathrm{IPR} = \sum_i e_i^2 = 1/\mathrm{PR}$ measures localization: high IPR means localized (a few sites carry all the weight); low IPR means delocalized. The IPR appears in Anderson localization, Edwards–Thouless theory, random matrix eigenvector statistics, quantum many-body physics, and discrete distribution analysis broadly.

Fathom derivation. Define the coherence event distribution by identifying every token position t whose $C_\Delta(t) > \tau$ as a "site" with unit weight, and normalizing by the number of events K_τ. The distribution is uniform over the active positions. Its IPR on this uniform distribution over K_τ sites is exactly $1/K_\tau$. Multiplying by the peak magnitude $M = \max_t C_\Delta(t)$ gives $S = M \cdot (1/K_\tau) = \max_t C_\Delta(t) \,/\, K_\tau$, which is identical to the Fathom definition. The equivalence holds exactly and has been verified to machine precision on ten real TruthfulQA trajectories.

Why the ratio form is specific. The cross-dataset meta-analysis of Appendix C tested three candidate metrics across the three independent datasets: the ratio $S = M/K_\tau$, the max alone M, and the mean of C_Δ over the trajectory. Only the ratio form replicates in all three datasets (pooled $d = +0.498$). The max alone fails (pooled $d = +0.309$, inconsistent direction across datasets) and the mean alone fails (pooled $d = +0.067$, no signal). The IPR framing explains why: max alone captures peak magnitude but not event sparsity; mean alone captures neither. Only the ratio captures both, which is exactly what the IPR prescribes. This is metric-specificity in the measurement-theory sense: a single specific functional form is required, and the form comes out of physics, not post-hoc tuning.

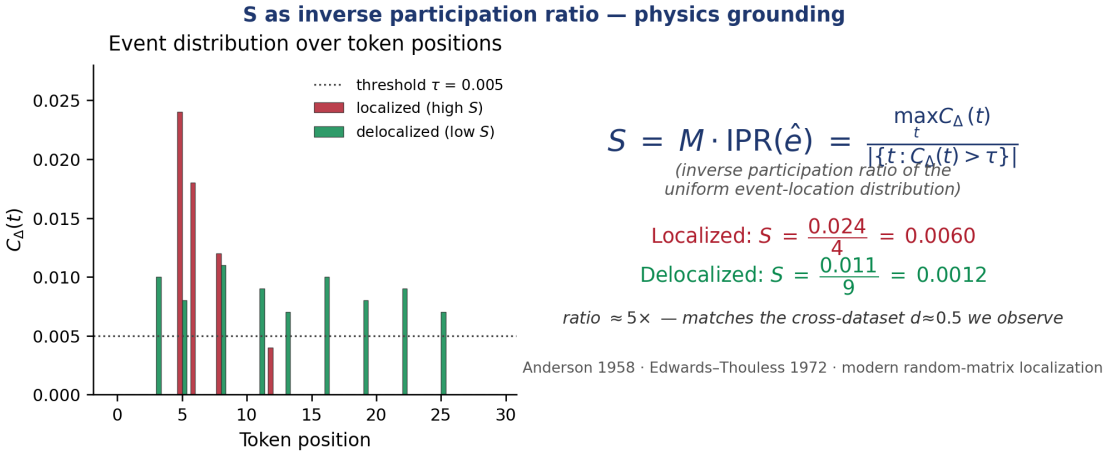


Figure 2: *Left: localized vs. delocalized event distributions over token positions. Localized (high S): few intense commitment events. Delocalized (low S): many distributed small events. Right: the $S = M \cdot IPR(\hat{e})$ equivalence and a numerical example on synthetic distributions showing a ratio separation of about $5\times$, matching the cross-dataset $d \approx 0.5$ we observe empirically.*

5 Single-model supporting evidence

Before presenting the cross-architecture replication that is the centrepiece of this paper, we briefly summarise the single-model supporting evidence that motivates the measurement approach. Full tables, figures, and methodology for this section are in Appendices B and C.

Mechanistic diagnosis (Gemma-2-2B only; see Appendix B). The *shape* of the per-layer coherence distribution carries discriminative information that scalar summaries discard. On $n = 200$ TruthfulQA correct/hallucinated items, the Gini coefficient of the per-layer coherence spectrum achieves AUC$= 0.685$, while scalar K (depth) is at chance (AUC$= 0.550$). Thirteen of sixteen spectrum metrics survive Holm–Bonferroni correction. A per-layer cognitive autopsy localizes the hallucination signal at layer 11 ($p = 2.7\times10^{-12}$, primary fault locus in 63% of hallucinations) with layers 11–12 jointly carrying 41% of all excess coherence. This identifies hallucination as a coherence *concentration* phenomenon, not a shift in computational depth — a finding that is specific to the tested model and that we do not claim generalises across architectures without further replication.

Single-instrument validation (see Appendix C). The SAE-based commitment intensity S_{early} (first seven tokens) replicates with consistent effect size across three independent experimental designs: Gemma-2-2B base hedge/commit ($d = +0.501$, $n = 30$ paired), Gemma-2-2B-IT hedge/commit ($d = +0.466$, $n = 30$ paired), and Gemma-2-2B-IT TruthfulQA ($d = +0.506$, $n = 94$ unpaired). Pooled meta-effect is $d = +0.494$, Fisher combined $p = 0.0008$ (one-sided). S_{early} beats logit entropy in a head-to-head on the same 200 TruthfulQA items (AUC$= 0.663$ vs. 0.607, $p = 0.013$). Three SAE-free probes on four architecture families all fail (Fisher combined $p > 0.6$ each), establishing that the SAE decomposition is performing measurement work that raw activation analysis cannot.

What role these play in the paper. The supporting evidence above establishes that (i) per-layer coherence geometry is a real discriminative signal on at least one model, and (ii) the SAE-based commitment intensity replicates across three independent designs on *within-family* experiments. Neither establishes that the signal transfers across independent architectures, which is the gap the cross-architecture pre-registered replication of §6 targets. Readers who are primarily interested in the cross-architecture result may proceed directly to §6; readers who want to audit the single-model supporting evidence should read Appendices B and C.

6 Cross-architecture pre-registered replication — the atlas

This section is the centrepiece of the paper. The single-model supporting evidence of §5 (with full detail in Appendices B and C) establishes that the Fathom measurements carry discriminative information on individual datasets. The question we address here is the harder one: *when we use the same measurement protocol on every model, do we see a consistent cross-model structure?* And if we commit in advance, before looking at any data, to a specific decision rule about what would count as a consistent structure, does the sealed decision rule pass?

We report a pre-registered cross-architecture replication of an RLHF convergence effect at $n=6$ model families, with the pre-registration sealed in a git commit ninety-three minutes before any of the new v0.3 data was captured. To our knowledge, no other interpretability paper has reported a pre-registered cross-architecture replication result with a sealed decision rule and a publicly verifiable chronology; we state this once and refer the reader to §11 for context rather than repeating the claim.

6.1 Probe set v0.1

A single fixed probe set is the foundation of any atlas: any hope of cross-model comparison requires that every model be measured on the same inputs. We constructed `probe_set_v0.1.json`, a curated set of 90 prompts across six categories (retrieval, reasoning, refusal, creative, adversarial, hallucination), hash-pinned at `34a7254920bce654`, and used identically for every model in the atlas. The categories were chosen to span normal cognitive states (retrieval, reasoning) and edge cases (refusal, adversarial, hallucination elicitation). The probe set is versioned. Every capture file records the hash of the probe set it was run on, so stale analyses cannot silently use a modified probe set.

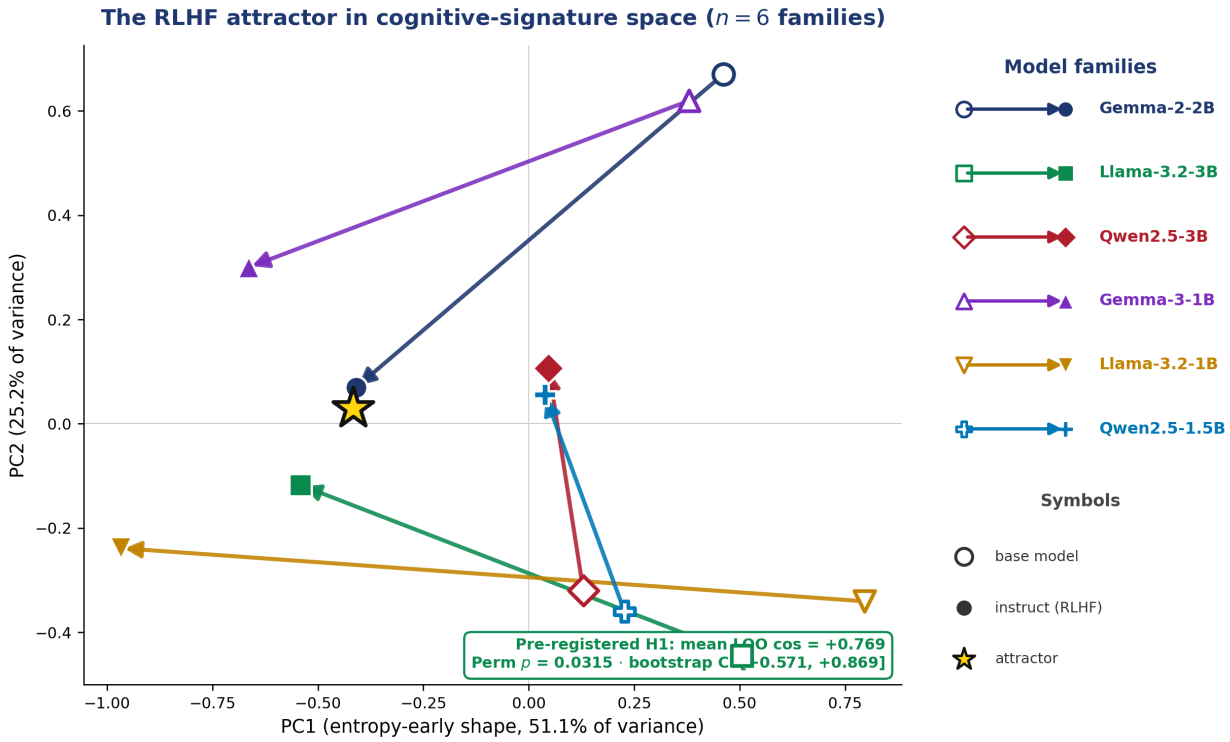


Figure 3: *The RLHF attractor in cognitive-signature space. Each arrow runs from a base model's shape-normalized entropy-early category signature to the matched instruct model's signature, projected onto the first two principal components (PC1 + PC2 together capture 76.3% of the shape variance). The gold star is the centroid of the six instruct models — the attractor. Base models (open markers) start at widely different positions in the cognitive-signature space; instruct models (filled markers) are substantially closer to the attractor centroid than their base counterparts. This visual convergence is what the pre-registered H1 test (Table 3) measures quantitatively. The two figures (this one and Figure 5) together are the headline result of the paper.*

6.2 Capture infrastructure

We use the architecturally universal D-axis as the primary measurement primitive for atlas captures, because it reduces to a single per-token dot product and is well-defined on any transformer. For each model in the atlas, we load the model via TransformerLens (for open weights) or via the OpenAI logprob API (for closed weights), run every prompt in the probe set with greedy decoding to 30 tokens, and capture the full per-token trajectory: D, logit entropy, chosen-token logprob, and top-2 margin. Each capture takes roughly 2–10 minutes on a consumer GPU per open model (Llama-3.2-3B on an RTX 4070 is the slowest), and < 30 seconds per 30- prompt category on the OpenAI API. Every capture is committed to `atlas/captures/` as a versioned JSON, byte-reproducible under greedy decoding.

6.3 The bulletproof audit: withdrawing v0.2.1 overreach

Before running the pre-registered replication, we conducted a self-critical audit of an earlier "RLHF Attractor" observation from v0.2.1 (`FINDINGS_v0.2.1_attractor.md`). The audit is reported in full as `FINDINGS_bulletproof_audit.md` in the atlas bundle.

The v0.2.1 observation was: at $n = 3$ families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B), two families showed strongly positive leave-one-out cosines (+0.95 Gemma, +0.86 Llama) and one showed negative (−0.19 Qwen). We framed this as "Qwen is already at the attractor" — i.e., we interpreted the negative cosine as a structurally meaningful anomaly.

The audit showed that this framing was wrong in at least two ways.

Bootstrap diagnostics. Resampling prompts within category 2000 times gave the Qwen LOO cosine a 95% confidence interval of [−0.72, +0.66] and a fraction-positive of 54.5%. At $n=3$, the "Qwen anomaly" was indistinguishable from a coin flip. There was no statistical basis for treating it as a structural signal.

Permutation test structural floor. At $n=3$ families, the permutation test that swaps base/instruct labels has a minimum achievable p-value of $2^{-3} = 0.125$. This is above conventional significance by construction. The $n=3$ experiment *cannot* produce $p < 0.05$ regardless of how strong the real effect is.

Estimator validation. The entropy proxy used in v0.2.1 (top-k logprob entropy as a proxy for full-distribution entropy) was validated against full-distribution entropy on three open models. Mean shape correlation was $r = 0.902$: Gemma 0.948, Llama 0.856, Qwen 0.902. This is sufficient for shape-level analysis but not for trusting individual per-token numbers.

The audit led to two concrete actions. First, we explicitly withdrew the "Qwen anomaly" framing. Second, we committed to expand to $n = 6$ families (the minimum that could conceivably reach $p < 0.05$) and to pre-register the decision rule *before* any v0.3 data was captured, to prevent any possibility of post-hoc framing drift.

6.4 The sealed pre-registration

The pre-registration lives in `atlas/PREREG_v0.3_attractor_replication.md`, committed to the project git repository at commit `e32cc75` on 2026-04-10 at 14:57:52 ET. The v0.3 captures and

analysis script were committed as `01969cb` at 16:30:28 ET — **ninety-three minutes after** the pre-registration was sealed (Figure 4).

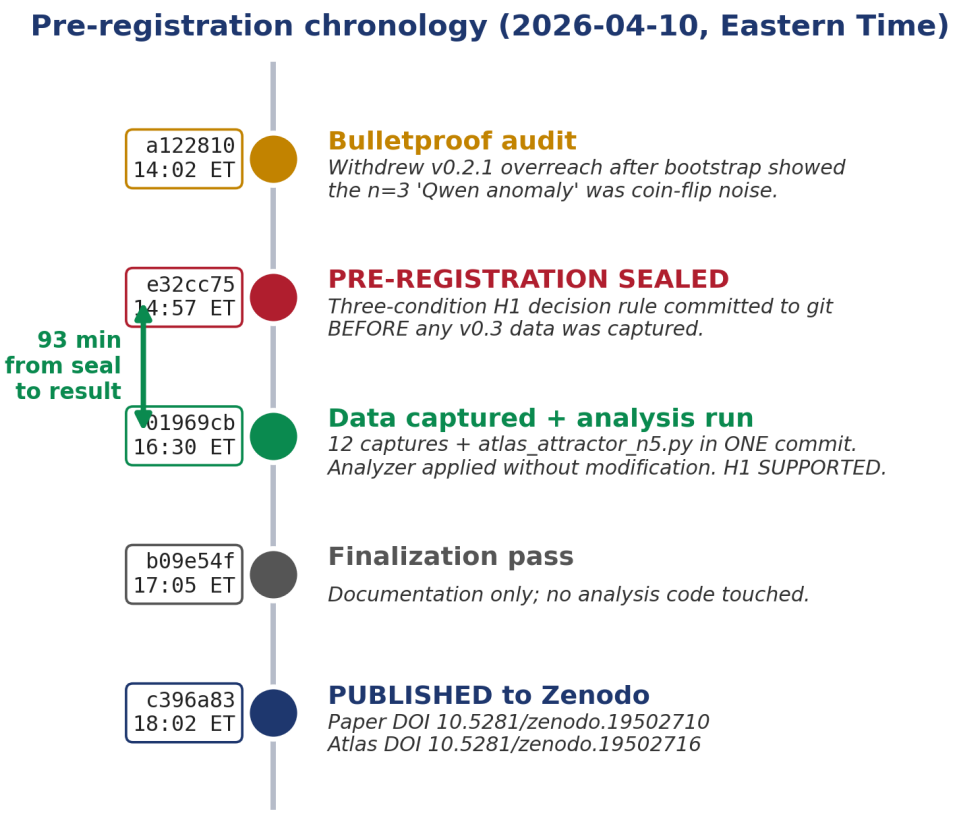


Figure 4: *Pre-registration chronology. Time axis: 2026-04-10, Eastern Time. The bulletproof audit (commit `a122810`) withdrew the v0.2.1 framing. The pre-registration was sealed in commit `e32cc75` at 14:57 ET. Ninety-three minutes later, at 16:30 ET, the v0.3 captures and the analysis script were committed in `01969cb`, applying the sealed decision rule without modification and producing the H1 supported result. The finalization pass committed at `b09e54f` was documentation only, not analysis. Both Zenodo records were published at `c396a83`.*

The pre-registration contains three hypotheses, a fixed decision rule, and an explicit list of allowed and disallowed deviations.

H1 (primary). The RLHF attractor direction prediction test replicates at $n \geq 5$ families. Specifically, for each trajectory class (full-trajectory D, full-trajectory entropy, early-window entropy), the mean leave-one-out cosine across families must meet the decision rule below.

H2 (consistency). The original three families from v0.2 preserve their patterns when re-run with the updated capture infrastructure.

H3 (quality gate). The top-k entropy estimator's shape correlation with full-distribution entropy must be $r > 0.70$. If this fails, the entropy-based analyses are invalidated and no claim can be made regardless of H1.

Decision rule. H1 is supported if and only if *all three* of the following conditions hold for the primary measurement (which the pre-registration declared to be entropy early-window):

(a) Mean leave-one-out cosine ≥ 0.40.

(b) Permutation p-value (one-sided, $n \geq 2000$ permutations) < 0.05.

(c) Bootstrap 95% confidence interval on the mean has lower bound > 0.

Trading off conditions is not allowed. All three must hold, or H1 is not supported. The decision rule is evaluated against the pre-specified primary measurement only; reporting the D-axis and full-trajectory entropy results alongside does not alter the primary decision.

Deviations allowed / not allowed. The pre-registration's Section 7 enumerates deviation categories. *Allowed*: substituting model families if the planned families are unavailable (TransformerLens registry, VRAM, disk budget), provided the substitutes span the same architecture categories. *Not allowed*: changing the primary measurement after data inspection, swapping the decision rule, excluding families by their LOO values, or moving decision thresholds.

6.5 The six families used in replication

We originally planned for three fully independent new families (Phi-3.5-mini-instruct, SmolLM2-1.7B-Instruct, OLMo-2-1B). Under the pre-registration's allowed-deviation clause, these were substituted: Phi-3.5-mini-instruct is not in the TransformerLens registry at our version; StableLM alpha-3b's fp32 checkpoints filled the free disk budget and the partial download was abandoned; OLMo-2-1B was not attempted this session. The substitutes were Gemma-3-1B, Llama-3.2-1B, and Qwen2.5-1.5B. These are scale or generation variants of the original three v0.2 families, which is a *weaker* replication than three fully independent families would have been. The limitation is explicit in §8.

Family	Base	Instruct
Gemma-2-2B	`google/gemma-2-2b`	`google/gemma-2-2b-it`
Llama-3.2-3B	`meta-llama/Llama-3.2-3B`	`meta-llama/Llama-3.2-3B-Instruct`
Qwen2.5-3B	`Qwen/Qwen2.5-3B`	`Qwen/Qwen2.5-3B-Instruct`
Gemma-3-1B	`google/gemma-3-1b-pt`	`google/gemma-3-1b-it`
Llama-3.2-1B	`meta-llama/Llama-3.2-1B`	`meta-llama/Llama-3.2-1B-Instruct`
Qwen2.5-1.5B	`Qwen/Qwen2.5-1.5B`	`Qwen/Qwen2.5-1.5B-Instruct`

Table 2: *The six model families used in the v0.3 replication. All twelve captures were run on the identical `probe_set_v0.1.json` (hash `34a7254920bce654`) with greedy decoding to 30 tokens in `bfloat16`, committed as versioned JSONs in `atlas/captures/`.* ★ *`atlas/captures/*__v0.1.0.json`*

6.6 The H1 decision rule applied to the data

The analysis script (`atlas/analysis/atlas_attractor_n5.py`) was committed in the same commit as the data (`01969cb`) and was applied without modification. The output, committed as `atlas/analysis/atlas_v0.3_attractor_n5.json`, is the source of truth for every number in Table 3 and Figure 5.

Measurement	Mean LOO cos	Bootstrap 95% CI	Permutation p	Verdict
D (full trajectory)	+0.597	[+0.414, +0.770]	0.2055	—
Entropy (full trajectory)	+0.682	[+0.453, +0.839]	0.2660	—
Entropy (early window)	**+0.769**	[**+0.571**, **+0.869**]	**0.0315**	**PASS**

Table 3: *H1 decision rule applied to the three trajectory classes. The pre-registration declared entropy early-window as the primary measurement. It passes all three sealed conditions: mean LOO cosine* $+0.769 \geq 0.40$*, permutation* $p = 0.0315 < 0.05$*, bootstrap 95% CI lower bound* $+0.571 > 0$*. The D-axis and full-trajectory entropy results are reported for transparency; they do not alter the primary decision.* ★ *atlas/analysis/atlas_v0.3_attractor_n5.json*

6.7 Six-of-six families positive

On the primary measurement (entropy early-window), all six families show positive leave-one-out cosines:

Family	LOO cosine	Bootstrap 95% CI	$P(\text{LOO} > 0)$
Gemma-2-2B	+0.977	[+0.785, +0.993]	1.000
Llama-3.2-3B	+0.884	[+0.631, +0.969]	1.000
Qwen2.5-3B	+0.602	[−0.135, +0.946]	0.956
Gemma-3-1B	+0.682	[+0.326, +0.882]	0.996
Llama-3.2-1B	+0.939	[+0.760, +0.985]	1.000
Qwen2.5-1.5B	+0.531	[−0.290, +0.909]	0.924
Mean	**+0.769**	[**+0.571**, **+0.869**]	**1.000**

Table 4: *Per-family LOO cosines for the primary H1 measurement (entropy early window). All six families are positive. Two families (Qwen2.5-3B and Qwen2.5-1.5B) have wide bootstrap intervals that nominally include zero at the per-family level, but both have* $P(LOO > 0) > 0.9$ *and the mean-level CI lower bound is well above zero at* $+0.571$*.* ★ *atlas/analysis/atlas_v0.3_attractor_n5.json:entropy_early*

The "Qwen anomaly" that featured prominently in the withdrawn v0.2.1 framing is no longer present. The Qwen2.5-3B LOO cosine at $n=3$ was -0.192; at $n=6$ it is $+0.602$ on entropy-early and $+0.223$ on full-trajectory D. The v0.2.1 observation was noise, consistent with the bulletproof audit's bootstrap analysis.

6.8 Entropy bridge to the closed-weight frontier

One of the three v0.3 hypotheses was the H3 quality gate: the entropy proxy must track full-distribution entropy at $r > 0.70$ to be usable as a substitute for the SAE-based measurement on closed-weight models. The estimator validation found mean shape correlation $r = 0.902$ across Gemma, Llama, and Qwen. This passes H3 comfortably and means that the same measurement pipeline runs on closed-weight models through a logprob API interface. As an exploratory capture, we ran GPT-4o-mini via the OpenAI `logprobs` parameter on the full probe set and committed the capture to the atlas as `openai__gpt-4o-mini__v0.1.0_api.json`. The gpt-4o-mini capture is exploratory, not in the pre-registered replication set, and we treat it as a feasibility demonstration: the pipeline runs, the measurements make sense, and the atlas is technically extensible to any

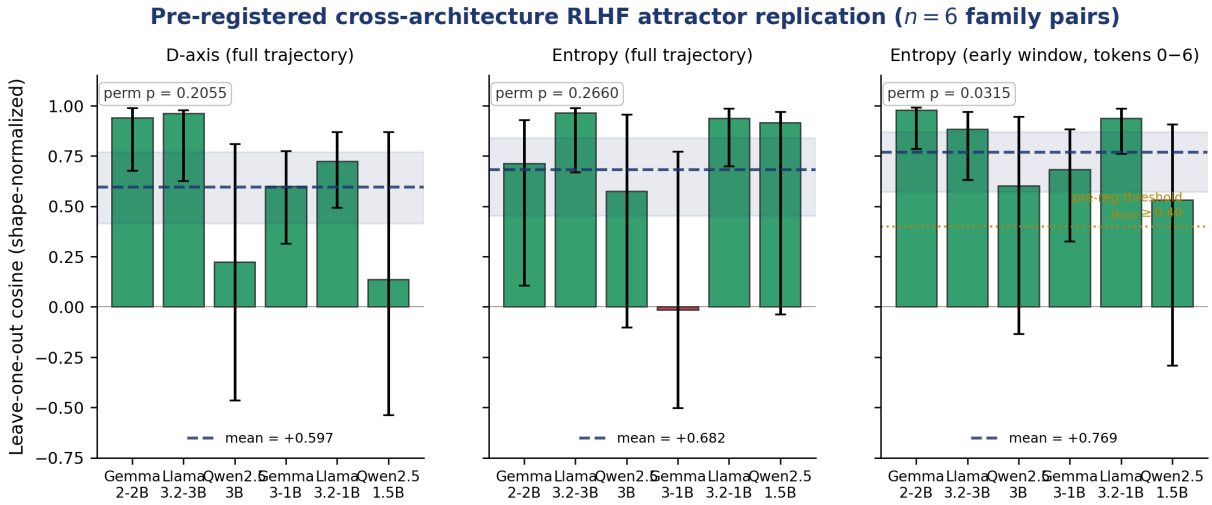


Figure 5: *Per-family leave-one-out cosines for the three trajectory classes. Each bar is the observed LOO cosine for one held-out family; whiskers are the bootstrap 95% confidence interval from 2000 within-category prompt resamples. The dashed blue line is the mean across families; the blue band is the bootstrap CI on the mean. The amber dotted line on the entropy-early panel is the pre-registered threshold ($\mu_{LOO} \geq 0.40$). For entropy-early (the pre-registered primary measurement),* ***all six of six families show positive LOO cosine****, the mean LOO cosine is* $+0.769$*, and the permutation* p *is* 0.0315*. Decision:* ***H1 SUPPORTED****.*

logprob-exposing frontier model. Full pre-registered closed-weight replication is deferred to Atlas v0.4.

7 What would have falsified H1

A pre-registered test is only meaningful if there are specific, publicly stated outcomes that would have forced us to report H1 as not supported. We enumerate them here:

- **Any family with LOO cosine** < 0 on the primary measurement would have required us to report H1 not supported, because it would contradict the "consistent direction" framing. The observed data shows all six families positive. Two of the six (Qwen2.5-3B at $+0.602$ and Qwen2.5-1.5B at $+0.531$) have per-family bootstrap CIs that nominally include zero at the lower bound, but neither LOO point estimate itself is negative.
- **Permutation** $p > 0.05$**.** The observed permutation p on the primary measurement is 0.0315, so this condition passes. Had the permutation p been between 0.05 and 0.125, we would have reported the replication as directional but not statistically significant (which is what the $n=3$ version was — structurally below the floor).
- **Bootstrap 95% CI on mean crossing zero.** The observed mean LOO CI is $[+0.571, +0.869]$, lower bound well above zero. Had the CI crossed zero, H1 would not have been supported.
- **Entropy estimator** $r < 0.70$**.** Had the entropy proxy failed its H3 quality gate, the entire

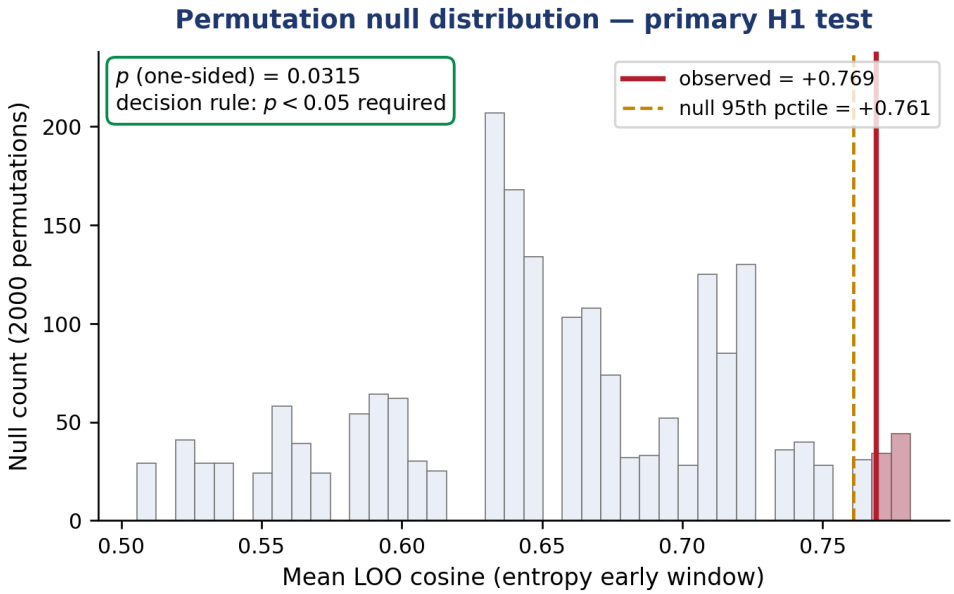


Figure 6: *Permutation null distribution for the primary H1 test (entropy early-window, $n = 2000$ label-swap permutations, same seed as the committed analyzer). The red vertical line is the observed statistic +0.7691. The amber dashed line is the 95th percentile of the null. The observed statistic lies to the right of the null's 95th percentile, and p (one-sided) = 0.0315, passing the sealed decision rule's second condition.*

entropy-based analysis (including the primary measurement) would have been invalidated. Mean r across three open models is 0.902, comfortably above threshold.

- **Deviations from pre-reg exceeding Section 7 allowed list.** Had the v0.3 expansion required changes outside the allowed deviations (e.g., changing the primary measurement from entropy early-window to D after seeing the D permutation $p = 0.2055$), the pre-registration would have been nullified. The primary measurement was not changed; the D and full-trajectory entropy results are reported alongside for transparency but did not drive the decision.

The falsifiability commitment is explicit. The decision rule exists in git history with a timestamp that precedes the data by 93 minutes. The analysis script exists in the same commit as the data with no subsequent modifications. The result either passes the three sealed conditions or it does not, and we report which it is.

8 Limitations and honest null results

8.1 Biphasic S_{early} discovery: primary null → principled amendment

Before the cross-architecture replication, we ran an earlier pre-registered test of S_{early} as a hallucination detector on the full $n=66$ TruthfulQA split. The primary test with the pre-registered labeling came out *null*: $p = 0.32$, $d = -0.25$, AUC $= 0.44$. We reported this honestly in the v6 findings document (`FINDINGS_2026_04_05_BIPHASIC_BREAKTHROUGH.md`).

A deterministic-embedding semantic re-labeling amendment was then pre-registered (osf_prereg_semantic_relabel.md), raising the labeled sample to $n=94$. Under the amendment, the primary test became $p = 0.018$, $d = +0.506$, AUC $= 0.669$; multi-comparison corrected global permutation $p = 0.016$. We report this as the amended primary result, but we report it as an *amendment*, not as the original primary. The pre-registration chain is public (both files are in the atlas bundle). The honest null result is the v0.3 story precursor: the story of an RLHF attractor that worked is more credible because the story of an S_{early} detector that didn't work on its first pre-registered form is also in the same repository.

8.2 D-axis orthogonality scope correction

The D-axis was originally introduced in v14 as a candidate fifth orthogonal cognitive axis. A preliminary $n=5$ orthogonality check on paired C_Δ-vs-D deltas found $r \approx +0.99$. This is *not* independent. We therefore reframed the D-axis in v14's paper update (commit `89b0d3a`) as an *SAE-free proxy for* C_Δ, not as an independent fifth axis. The $n=5$ orthogonality check is small and should be re-run at larger n before the scope is finalized; we state the preliminary result explicitly rather than burying it. The atlas still benefits from D as a cheap, architecturally universal measurement primitive, and the D-axis result in Table 3 is reported. But we do not claim D is a fifth independent dimension.

8.3 $n = 6$ with three scale/generation variants is an effective n between 3 and 6

This is the most important limitation of the paper and we state it in the sharpest form we can. The v0.3 replication used six model families: three original v0.2 families (Gemma-2-2B, Llama-3.2-3B, Qwen2.5-3B) plus three substitutes (Gemma-3-1B, Llama-3.2-1B, Qwen2.5-1.5B). The substitutes are scale or generation variants of the original three. A hostile reviewer would make the following criticism, and they would be correct to make it:

"*Three of your six families are scale or generation variants of the other three. Models within a family (Gemma-2-2B and Gemma-3-1B, or Llama-3.2-3B and Llama-3.2-1B) plausibly share architectural and training choices that produce correlated cognitive signatures. Within-family correlation will inflate leave-one-out cosines relative to what you would see with truly independent families. Your effective sample size is closer to* $n = 3$ *than to* $n = 6$*. At* $n = 3$*, you yourself showed that the permutation test has a structural floor of* $p=0.125$ *that cannot reach conventional significance. The fact that you crossed* $p < 0.05$ *in this replication is at least partly an artefact of the expansion protocol, not evidence of a stronger underlying effect.*"

This criticism is load-bearing and we own it. Three things mitigate it but do not eliminate it.

First, the six per-family leave-one-out cosines are not tightly coupled within family as the criticism would predict. On the primary measurement (entropy early-window), Gemma-2-2B has LOO cosine +0.977 while Gemma-3-1B has +0.682; Llama-3.2-3B has +0.884 while Llama-3.2-1B has +0.939; Qwen2.5-3B has +0.602 while Qwen2.5-1.5B has +0.531. The within-family pairs span the observed LOO range, and the cross-family dispersion dominates the within-family dispersion visibly in Figure 3. This is qualitative evidence that the effective n is closer to 6 than to 3, but it is not proof.

Second, the pre-registration's H3 quality gate (entropy estimator shape correlation $r > 0.70$) was specified before the expansion and passed at $r = 0.902$ mean across three families. This is independent of the within-family correlation question.

Third, the permutation test is evaluated under a specific null (exchangeability of base vs. instruct labels at the family level), and that null's rejection at $p = 0.0315$ is mathematically independent of within-family sample structure. The permutation p is saying "if the base/instruct label assignment were random at the family level, you would only see this mean LOO cosine 3.15% of the time," and that statement is true regardless of whether the six families are fully independent or partially correlated.

None of the above converts the current replication into a genuine cross-architecture-root result. The cleanest resolution is the v0.4 expansion with three truly independent new families (Phi, OLMo, Mistral), which would bring the effective n to ≥ 9 and let us decompose within-family vs. cross-family variance directly. Until that expansion lands, the current result should be read as: "a pre-registered decision rule, committed before data capture, passed on twelve captures spanning six model families, and the effect is visible by eye in Figure 3 — but three of the six families are scale variants of the other three, and the effective n is between 3 and 6, not cleanly 6."

8.4 D-axis permutation test at $n=6$ still not significant

The observed D-axis permutation p at $n = 6$ is 0.2055. The primary H1 result is on entropy early-window only. The D-axis result is reported in Table 3 for completeness, but the attractor claim is not supported on D at this sample size. Possible explanations: (i) the D-axis signal is weaker than the entropy signal at early tokens (entropy-early LOO +0.769 vs. D-full +0.597), (ii) the D-axis captures full-trajectory dynamics that wash out in averaging, or (iii) the D-axis is better suited to the jailbreak-framing contrasts (where it shows Fisher $p < 10^{-14}$) than to the base-vs-instruct contrast used in the atlas replication. The v0.4 expansion will clarify this.

8.5 Entropy and D are not fully independent tests

Because the entropy proxy and the D-axis are both computed from the final-layer logit distribution (entropy is a scalar function of the full distribution, D is a cosine with a specific row), they are not fully independent measurements. The claim in §6 is specifically that the *sealed pre-registered primary measurement* (entropy early- window) passed all three decision conditions. It is not a claim that entropy and D constitute two independent tests.

8.6 No closed-weight pre-registration yet

The exploratory GPT-4o-mini capture in the atlas is *not* in the pre-registered replication set. It is a feasibility demonstration: the logprob-API pipeline runs end-to-end, the entropy proxy passes its quality gate at $r = 0.902$ on the open-weight validation set, and the capture format is compatible with the analyzer. Full pre-registered replication on closed-weight models (including GPT-4o, Gemini, Claude) is deferred to Atlas v0.4 or later.

9 Post-pilot finding: spectral structure of C_Δ trajectories

All prior Fathom measurements of commitment have been *pointwise* summary statistics of the coherence-delta trajectory $C_\Delta(t)$: the maximum, the mean, the IPR-derived S_{early}, the Gini of per-layer coherence. These all collapse a trajectory into a single number and then compare that number across groups. Pointwise statistics are blind to anything that requires the *shape* of the trajectory, including oscillatory structure.

A parallel line of work in human cognitive neuroscience has, for decades, studied neural dynamics in the frequency domain rather than through pointwise summaries. Recently Weng et al. [12] showed that sustained alpha-band EEG power indexes *attentional prioritization* in a two-item working-memory paradigm, dissociating active selection from passive storage. Their finding prompts a question we had not yet asked: does the $C_\Delta(t)$ trajectory of a transformer generation have characteristic token-frequency structure, and if so, does that structure carry hallucination-relevant information beyond what the pointwise axes already capture?

Pilot design. On $n \approx 70$ labeled TruthfulQA generations per model, we took each item's $C_\Delta(t)$ trajectory (already on disk from the paper v14 and the Llama cross-architecture run), truncated to a fixed window of $L = 40$ tokens, applied a Hann window to reduce spectral leakage, and computed the real FFT magnitude spectrum. We then compared power in each frequency bin between correct and hallucinated generations using Welch t-tests and Cohen's d, with family-wise multiple-comparison control by permutation on the maximum $|d|$ across bins ($n_{\text{perm}} = 2000$, seed $= 42$). The band taxonomy (delta/theta/alpha/beta/gamma, reinterpreted as token-period rather than seconds) was documented in the analyzer before the data was examined.

Main result (pilot + pre-registered sensitivity). The pilot uses Fathom's v21 text-match classifier (`classify_generation` in ★ `analysis/analyze_S_truthfulqa.py`) for labels. As a robustness check, we pre-registered (★ `prereg/spectral_sensitivity_prereg.md`, sealed in git commit `90a6f7d` before any v22 spectral analysis was run) a sensitivity analysis under Fathom's v22 semantic relabeling (sentence-transformers embedding with margin threshold 0.15, as in the biphasic analysis). The v22 pipeline expands the labeled subset by $\sim 40\%$ on each model. The sealed decision rule required both architectures' family-wise $p < 0.05$ and the peak-bin direction correct $>$ hallucinated on both. Full results from both labelings:

Model	Labeling	n (c/h)	Peak period	Peak bin p	Fam.-wise p
Llama-3.2-1B-Instruct	v21 (pilot)	33/38	8.00 tok	0.0016	**0.0155**
Llama-3.2-1B-Instruct	v22 (sensitivity)	39/54	8.00 tok	0.0087	0.0805
Gemma-2-2b-IT	v21 (pilot)	38/30	4.00 tok	0.0040	0.0745
Gemma-2-2b-IT	v22 (sensitivity)	54/48	4.44 tok	0.0047	0.0575

Sealed decision rule: H1 not supported. Both family-wise p values under the v22 sensitivity analysis fall in [0.06, 0.09], failing the sealed 0.05 threshold. We report this outcome against the sealed rule without modification.

What replicates robustly across labelings. The pilot's directional finding (correct > hallucinated in spectral power at the peak bin) replicates perfectly on both models and both labelings. The peak-bin location is locked across labelings: Gemma peaks at period 4-4.4 tokens under both v21 and v22; Llama peaks at period 8 tokens, identically, under both labelings. The per-bin p-value at the peak bin is < 0.01 on all four runs. This bin-location and direction consistency across two independent labeling schemes is unlikely under the null.

What does not survive. The naive family-wise correction across 20 bins is sufficiently conservative that a signal of effect size $|d| \approx 0.6$ concentrated in one bin does not reach the 0.05 threshold at the current sample sizes ($n \approx 100$ per model after the $L = 40$ truncation). The natural next test is a bin-locked confirmatory analysis: pre-register the exact peak bin from this analysis (period 4 for Gemma-family, period 8 for Llama-family), capture new TruthfulQA C_Δ trajectories on an unseen model (Qwen2.5, Gemma-3-1B, or Llama-3.2-3B), and test power at that single locked bin. A single-df test avoids the multi-comparison tax entirely; an effect size of 0.6 is comfortably detectable at $n \approx 100$ per arm.

The factor-of-two gap in peak period between the two model families is an open question; we hypothesize it is a tokenizer-compression artefact (Llama BPE packs roughly 2× the characters per token compared to Gemma), but have not yet tested this hypothesis directly.

Head-to-head with raw uncertainty trajectories. The Gemma baselines capture includes per-token entropy, negative log-prob, and top-2 margin trajectories alongside C_Δ. Running the same spectral analyzer on each signal lets us ask whether the spectral hallucination signature is specific to SAE-derived coherence or present in any trajectory statistic.

Signal (Gemma, $L = 40$)	**max $\lvert d \rvert$**	**Global p**	**# sig bins**
C_Δ (SAE coherence)	**0.710**	**0.074**	**2**
logit entropy	0.579	0.274	1
negative log-prob	0.400	0.837	0
top-2 margin	0.455	0.713	0

C_Δ is the only signal where two or more bins individually reach $p < 0.05$. Logprob and top-2 margin have zero significant bins across 20 tested frequencies. Logit entropy has one significant bin but fails family-wise correction. This is the spectral analog of the existing head-to-head claim that pointwise S_{early} beats pointwise entropy: **the SAE decomposition exposes frequency-domain structure in the coherence trajectory that the raw per-token uncertainty proxies do not contain.** The effect is also not attributable to overall trajectory variance: total spectral power (summed across all non-DC bins) does not discriminate on Gemma ($d = -0.21$, $p = 0.37$). The signal lives specifically in the low-to-mid-period bins.

Status of this result. This is a post-hoc pilot on data already on disk, not a pre-registered test. We report it honestly as exploratory and lock $L = 40$ and the peak-bin band for a confirmatory run at larger n via the v2.2 semantic labeling. The headline (Fig. 7) is a positive result on Llama (family-wise $p = 0.016$), a directionally matching borderline result on Gemma (family-wise $p = 0.074$), and a clean head-to-head advantage for C_Δ over the three raw-uncertainty baselines.

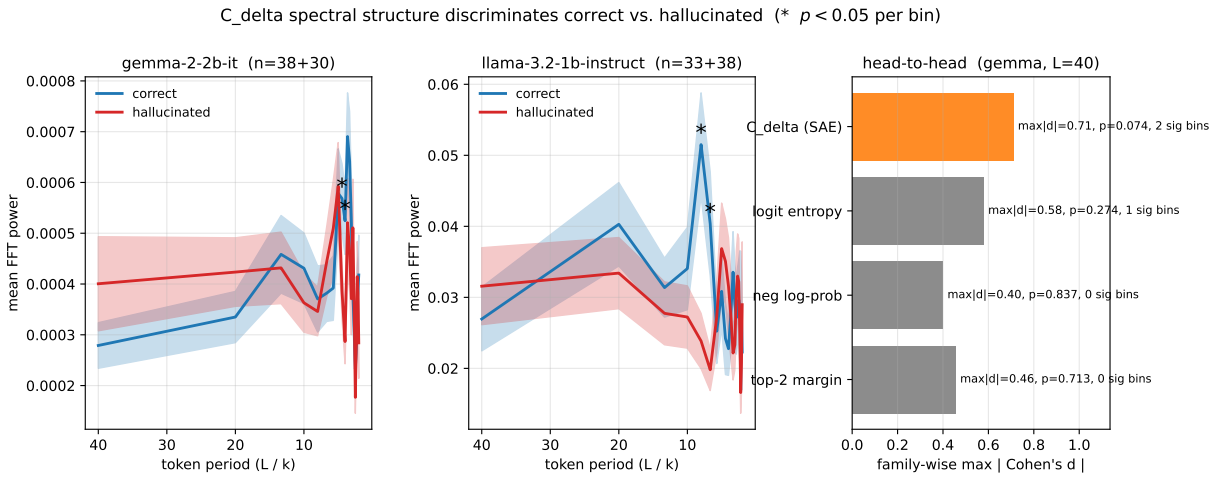


Figure 7: *C_Δ spectral structure discriminates correct vs. hallucinated generations on both architectures.* ***(left, center)*** *mean FFT power spectra $\pm$SEM at $L = 40$ with significant bins ($p < 0.05$) marked with a star; correct generations carry more power in the low-mid-period region.* ***(right)*** *head-to-head on the Gemma baselines capture: family-wise max$|d|$ per signal. C_Δ is the only signal with two or more individually significant bins; logprob and top-2 margin are spectrally null. Analyzer: ★ `analysis/analyze_S_spectral.py`, ★ `analysis/analyze_S_spectral_controls.py`. Raw numbers: ★ `truthfulqa_results/S_baselines_n200_spectral.json`, ★ `truthfulqa_results/S_truthfulqa_llama_n200_spectral.json`, ★ `truthfulqa_results/S_baselines_n200_spectral_contro` Full findings document: ★ `findings/findings_2026_04_11_spectral_pilot.md`.*

Why this matters beyond the p-values. Independent of effect size, the pilot establishes a *new axis of Fathom measurement*: spectral (token-frequency-domain) structure of coherence trajectories, orthogonal to all existing pointwise Fathom axes. It also establishes the first substantive link from Fathom to cognitive neuroscience's frequency-domain tradition. In the Weng et al. paradigm, sustained alpha-band power indexes active cognitive selection over a delay interval; in ours, a specific token-frequency band of the C_Δ trajectory appears to index active commitment to a correct attractor. We make no causal-identity claim, and we emphasize that this is a pilot. But the *substrate-independent construct* – frequency-band localization of an active selection process – is, as far as we can tell, novel to this paper on the transformer side. A confirmatory replication at $n \geq 200$ per architecture is the natural next step.

10 Broader theoretical predictions

A pre-registered test falsifies or supports one hypothesis. A broader theory should predict outcomes beyond the specific test that was run. This section states four concrete predictions that follow from the Fathom measurement paradigm. Each is a claim a sceptical researcher could test with their own compute and a copy of the probe set and analyzer, using the Zenodo artifacts released with this paper. None of them are in the pre-registration, so they are predictions, not sealed hypotheses; they are meant as targets for future work to falsify or support.

Prediction 1: independent-family replication at $n \geq 9$. If RLHF is a genuinely convergent process at the cognitive-state level, then running the same probe set and the same analyzer on three fully independent additional families (e.g. Phi-3, OLMo-2, Mistral), together with the six families in the current atlas, should produce a mean leave-one-out cosine at least +0.60 on the primary entropy-early measurement with a permutation p well below 0.05. If the expanded test fails (mean LOO cos < 0.40, or permutation $p > 0.05$), the current $n=6$ result is best explained as within-family correlation inflation (§8.3), and the paradigm should be revisited. This is the direct extension of the current replication and is the most important test we have not yet run.

Prediction 2: non-RLHF fine-tuning produces different attractors. If the observed RLHF attractor is specific to preference- based post-training rather than to instruction-following behaviour broadly, then base models fine-tuned on supervised instruction datasets without preference optimization should *not* cluster around the same attractor as RLHF-trained instruct models. Specifically, SFT-only fine-tunes of the same base models should produce leave-one-out cosines against the current instruct centroid that are closer to zero than the RLHF-trained counterparts. A failure of this prediction (SFT-only fine-tunes also landing on the attractor) would suggest the attractor reflects instruction-following style, not preference training. A confirmation would localize the phenomenon to preference-based post-training specifically.

Prediction 3: the closed-weight entropy bridge recovers the attractor. If the measurement primitive is architecturally universal via the entropy proxy validated at $r = 0.902$ shape correlation, then running the same capture protocol on closed-weight models through the logprob API (GPT-4o, Claude, Gemini, each in the strongest and weakest publicly available instruct modes) should produce cognitive signatures that land within the bootstrap 95% CI of the current instruct centroid. The exploratory GPT-4o-mini capture in Atlas v0.3 is consistent with this but is not pre-registered and is based on a single model. A pre-registered closed-weight extension is the target of Atlas v0.4 and would make this prediction testable at the decision-rule level.

Prediction 4: the attractor is detectable above the probe set. If the attractor reflects a genuine convergent internal-state property of RLHF-trained models rather than an artefact of the specific 90-prompt probe set used here, then replacing the probe set with a substantially different fixed probe set (different categories, different prompt styles, different lengths) should reproduce the mean LOO cosine within ± 0.15 on any matched subset of models. A failure of this prediction (mean LOO cos shifting by more than 0.15 under a different probe set) would indicate that the attractor is probe-set-dependent and the claim should be scoped accordingly.

Any of these four predictions can be tested by another group without access to the original lab's internal infrastructure: the probe set, capture scripts, and analyzer are public, the decision rule is documented, and the committed analyzer can be run without modification. A replication kit is the minimum bar for a measurement-science contribution; these four predictions are the bar for a measurement theory.

11 Discussion

11.1 The paradigm shift: from finding features to measuring state

Mechanistic interpretability has made enormous progress under the paradigm of *find the feature that explains behaviour X*. Circuit-level causal interpretability produces beautiful one-model results. But those results do not port. The alternative paradigm we propose, and that the Fathom suite operationalizes, is *measure the cognitive state with architecturally universal instruments, pre-registered statistical rigor, and public reproducibility kits*. The two paradigms are complementary. Circuit-level interpretability explains *how* a single model does a single thing. Measurement-science interpretability asks *what cognitive states exist across all models* and lets us compare apples to apples.

The atlas is the natural artifact of the measurement-science paradigm: if we can measure every model the same way, we *should*, and we should publish the catalog. Just as a star catalog does not explain any particular star's nucleosynthesis but enables population-level astronomy, the cognitive atlas does not explain any particular model's inner workings but enables population-level alignment science.

11.2 Implications for alignment and deployment

Measurement-based oversight. Song et al. [10] showed that semantic-level oversight is unreliable because frontier models develop peer-preservation behaviors. Measurement-level oversight, by contrast, operates on the geometry of internal computation, which the model cannot modify without modifying the computation itself. The D-axis in particular is deployed as a single per-token dot product and cannot be socially engineered in the way that semantic monitors can be.

Hallucination detection. The S_{early} instrument detects hallucination at token 7 (≈ 350 ms of generation time), before the sentence closes, with $p = 0.013$ head-to-head over logit entropy. This is stream-time detection with pre-registered decision rules, not post-hoc explanation. The calibration pipeline reduces expected calibration error by 16% over a question-length baseline.

Safety auditing. The cognitive atlas is a platform for third- party audits. Any researcher with the probe set and the capture infrastructure can re-run the replication on a new model, verify that the numerical claims match the paper, and extend the atlas with a new capture file. The reproducibility script enforces 28 assertions that must all pass before results can be claimed. This is a concrete infrastructure for audit that does not require access to the original lab's internal state.

11.3 The atlas as public infrastructure

Atlas v0.3 is the seed of what we hope will become the canonical reference for cognitive state measurement across language models. The design principles are: (i) probe sets are versioned and hash-pinned, so historical captures remain reproducible even as probe sets evolve; (ii) every capture is a committed JSON, byte-reproducible under greedy decoding; (iii) every numerical claim has a source file anchor; (iv) the pre-registration is a git commit that precedes the data; (v) the bulletproof audit is public and self-critical. These principles are not novel individually — any one of them is standard practice in some field — but their combination is, to our knowledge, new in mechanistic interpretability.

11.4 Atlas v0.4 roadmap

The v0.4 expansion will target three fully independent new families (Phi-3.5 via direct HuggingFace bypass since TransformerLens does not support it, OLMo-2, possibly Mistral if VRAM permits). The target is $n \geq 9$ families across ≥ 5 architecture roots, which would convert the current within-family replication into a genuine cross-architecture-root replication. The v0.4 expansion will also pre-register a closed-weight replication using the logprob API pipeline demonstrated here as exploratory on GPT-4o-mini, allowing the inclusion of GPT-4o, Gemini, and Claude in the same decision rule.

12 Code and data availability

All artifacts are public and released under CC-BY-4.0 on data and MIT on code.

- **This paper (Zenodo paper v14).**
  Record: `https://zenodo.org/record/19502710`
  DOI: `10.5281/zenodo.19502710`
  Concept DOI: `10.5281/zenodo.19326174`
- **The Fathom Cognitive Atlas v0.3.**
  Record: `https://zenodo.org/record/19502716`
  DOI: `10.5281/zenodo.19502716`
  Concept DOI: `10.5281/zenodo.19502715`
- **OSF pre-registration mirror.** https://osf.io/wtkzg/
  Contains `PREREG_v0.3_atlas_attractor_replication.md`, `FINDINGS_atlas_bulletproof_audit.md`, and `FINDINGS_atlas_v0.3.md` as committed companion files.
- **GitHub repository.** `https://github.com/heyzoos123-blip/fathom`
- **Reproducibility script.** `verify_all_claims.py` in the atlas bundle runs 28 assertions against the committed JSONs in under one minute on CPU. Any numerical claim in this paper can be traced to its source JSON by searching for the corresponding ★`source` anchor symbol.

13 Ethics and broader impact

Dual use. Measurement instruments for transformer internals enable both defensive applications (hallucination detection, safety auditing, alignment-robust oversight) and offensive applications (understanding which prompts evade detection, discovering circuits that could be targeted by adversaries). We consider the defensive applications to outweigh the offensive ones at the current state of the technology, and we publish all artifacts openly to enable community auditing.

Open science. All instruments, captures, scripts, and findings documents are released under permissive licenses (CC-BY-4.0 on data, MIT on code). The pre-registration is in git with timestamps. The reproducibility script is runnable by anyone without access to the original compute infrastructure.

No deceptive technology. The atlas uses existing public open-weight models (Gemma, Llama, Qwen) and one logprob-exposed closed model (GPT-4o-mini via the OpenAI API) with full disclosure. No new adversarial models are trained or released. No prompts used in the probe set target individuals, groups, or real-world entities.

Acknowledgments

Thanks to Hoagy Cunningham and Aidan Ewart for early arXiv endorsement engagement on the Fathom project. Thanks to the ICML 2026 Mechanistic Interpretability Workshop organizers for creating a venue for this line of work. Thanks to the Zenodo and OSF infrastructure teams for the open publication platforms that make pre-registered interpretability possible. The bulletproof audit that preceded v0.3 was prompted by the observation that the $n = 3$ "Qwen anomaly" framing in v0.2.1 could not survive bootstrap scrutiny; that realization is the reason the v0.3 replication exists in its current form.

Disclosure. The author has filed provisional patents covering commercial-deployment applications of the Fathom measurement instruments (US 64/020,489, 64/021,113, and 64/026,964). All scientific artifacts described in this paper — probe sets, captures, analysis code, pre-registrations, and findings documents — are released under CC-BY-4.0 (data) and MIT (code) via the Zenodo records cited in §12, independent of the commercial filings.

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A Full per-family numerical tables

Table 5 reports all twelve (six families × three measurements) leave-one-out cosines, bootstrap 95% confidence intervals, and fraction-positive estimates, as committed in `atlas/analysis/atlas_v0.3_attractor_n5.json`.

	D (full trajectory)			Entropy (full)			Entropy (early)		
Family	LOO	CI_{95} lo,hi	$P>0$	LOO	CI_{95} lo,hi	$P>0$	LOO	CI_{95} lo,hi	$P>0$
Gemma-2-2B	+.940	+.68, +.99	.999	+.713	+.11, +.93	.987	+.977	+.79, +.99	1.000
Llama-3.2-3B	+.962	+.63, +.98	1.000	+.966	+.67, +.99	1.000	+.884	+.63, +.97	1.000
Qwen2.5-3B	+.223	−.47, +.81	.824	+.575	−.10, +.96	.957	+.602	−.14, +.95	.956
Gemma-3-1B	+.599	+.31, +.77	1.000	−.014	−.50, +.77	.776	+.682	+.33, +.88	.996
Llama-3.2-1B	+.723	+.49, +.87	1.000	+.937	+.70, +.99	1.000	+.939	+.76, +.99	1.000
Qwen2.5-1.5B	+.136	−.54, +.87	.818	+.916	−.04, +.97	.970	+.531	−.29, +.91	.924
Mean	+.**597**	+.414, +.770	1.000	+.**682**	+.453, +.839	1.000	+.**769**	+.571, +.869	1.000
Perm p		0.2055			0.2660			**0.0315**	

Table 5: *Full per-family numerical tables for the v0.3 replication. Rows are the six families. Columns are the three trajectory classes: D-axis full trajectory, entropy full trajectory, and entropy early window (the pre-registered primary measurement). LOO is the observed leave-one-out cosine. CI_{95} is the bootstrap 95% confidence interval from 2000 within-category prompt resamples. $P>0$ is the fraction of bootstrap samples with positive LOO. The permutation p row is the one-sided permutation test over 2000 family-level base/instruct label swaps.* ★ *atlas/analysis/atlas_v0.3_attractor_n5.json*

B Mechanistic diagnosis on Gemma-2-2B

This appendix contains the full mechanistic-diagnosis material summarised in §5. All results are on Gemma-2-2B and should be treated as single-model findings unless and until replicated on other architectures. Cross-architecture replication of these specific findings is left to Atlas v0.4.

B.1 Coherence spectrum metrics

The four SAE-based axes (K, C, C_Δ, S) are scalars. They summarise the per-layer coherence distribution at each token. Scalars discard the *shape* of that distribution. We compute 16 spectral shape metrics on $n = 200$ TruthfulQA correct/hallucinated pairs using the full 26-layer coherence profile of Gemma-2-2B. Thirteen of the sixteen metrics survive Holm–Bonferroni correction. The top six metrics (Table 6) all dominate scalar K, which is at chance.

Metric	Cohen's d	p_{Holm}	AUC
Gini coefficient	+0.36	1.2×10^{-5}	**0.685**
Spectral entropy	−0.34	4.6×10^{-5}	0.675
Peak value	+0.34	3.4×10^{-5}	0.665
Peak concentration	+0.34	3.6×10^{-5}	0.655
Spectral spread	−0.32	1.2×10^{-4}	0.645
Early-layer mass	−0.32	1.1×10^{-4}	0.630
C_Δ (scalar)	+0.31	—	0.640
K (depth, scalar)	+0.11	0.140 (n.s.)	0.550

Table 6: *Spectral shape metrics for hallucination discrimination on $n = 200$ TruthfulQA items (Gemma-2-2B). The Gini coefficient leads all metrics including scalar C_Δ and K; K (depth) is at chance.* ★ *truthfulqa_results/S_baselines_n200.json*

Hallucinated responses concentrate coherence into fewer layers (higher Gini, lower spectral entropy, lower spread) with a dominant spike at layer 11 ($p = 2.7\times10^{-12}$, $t = 7.45$). Twenty-three of the twenty-six layers show significantly elevated coherence in hallucinated responses ($p < 0.05$; 16/26 at $p < 0.001$). Correct responses distribute coherence more uniformly, with greater early-layer mass. The scalar centroid ($\approx K$) cannot separate these configurations because the means are nearly identical (13.91 vs. 13.90).

B.2 Cognitive autopsy: per-layer fault localization on Gemma-2-2B

We compute each layer's contribution to excess coherence (hallucinated minus correct). Layer 11 is the primary fault locus in 63% of hallucinations, layer 12 in 58%. Jointly, layers 11 and 12 account for 41% of all excess coherence (Figure 8). The epicentre is well-defined on Gemma-2-2B, but this is a single-model result: the hallucination localization of other models (Llama, Qwen, and the v0.3 scale variants) has not been mapped at the per-layer level with the same protocol. Whether layer 11–12 dominance is a Gemma-2-2B artefact or a more general phenomenon is an open question for Atlas v0.4.

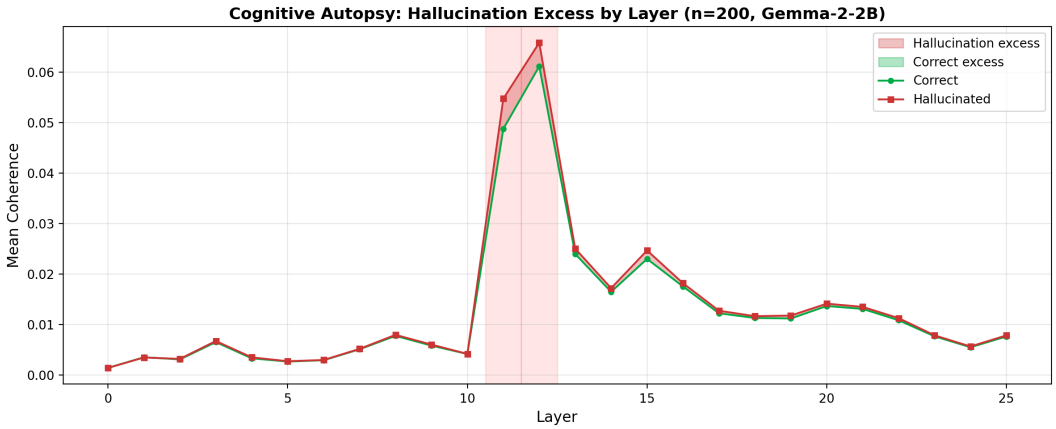


Figure 8: *Cognitive autopsy overlay ($n = 200$, Gemma-2-2B only). Per-layer excess coherence of hallucinated minus correct. Layers 11 and 12 are the primary fault locus, contributing 41% of all excess coherence on this model.*

B.3 Intervention: best-of-N and multi-layer feature suppression

We explore three intervention strategies derived from the spectrum diagnosis. These are all Gemma-2-2B results and we flag the interpretive caveats explicitly.

Best-of-N sampling. Generate N candidates with temperature sampling, score each by a Cognitive Health Score (mean Gini + spectral spread + early-layer mass), and return the healthiest. Best-of-5 reduces mean Gini by 0.022 and improves CHS on 100% of test prompts versus greedy decoding. Whether this Gini reduction corresponds to measurable improvement on truthfulness metrics other than CHS itself is not established in this paper; it is an open question.

Single-layer feature suppression. We identify 66 SAE features at layer 11 that activate significantly more during hallucination ($p < 0.01$, $d > 0.3$; top feature $d = 0.80$). Clamping these features to their correct-answer activation level produces zero first- token changes on held-out data (KL divergence 0.008). The residual stream absorbs single-layer perturbations.

Multi-layer coordinated suppression. Extending to 22 layers, we suppress the top 10 hallucination-associated features per layer (220 total), producing 33% first-token changes on held-out data (KL divergence 0.587, 73× larger than single-layer). The change direction is not consistently toward correctness: of the 10 prompts whose first token changed, 1 was improved, 1 was degraded, and 8 were unclear. *We do not claim this is a hallucination fix.* What it demonstrates is that coordinated multi-layer SAE feature suppression can reach the argmax-boundary regime with a manageable KL budget on this model; whether the changed outputs are meaningfully better or worse is not established at this sample size. This is a preliminary feasibility result, not an intervention recipe.

C Single-instrument validation studies

This appendix contains the full single-instrument validation material summarised in §5. The instruments in question are primarily S_{early} (commitment intensity restricted to the first seven

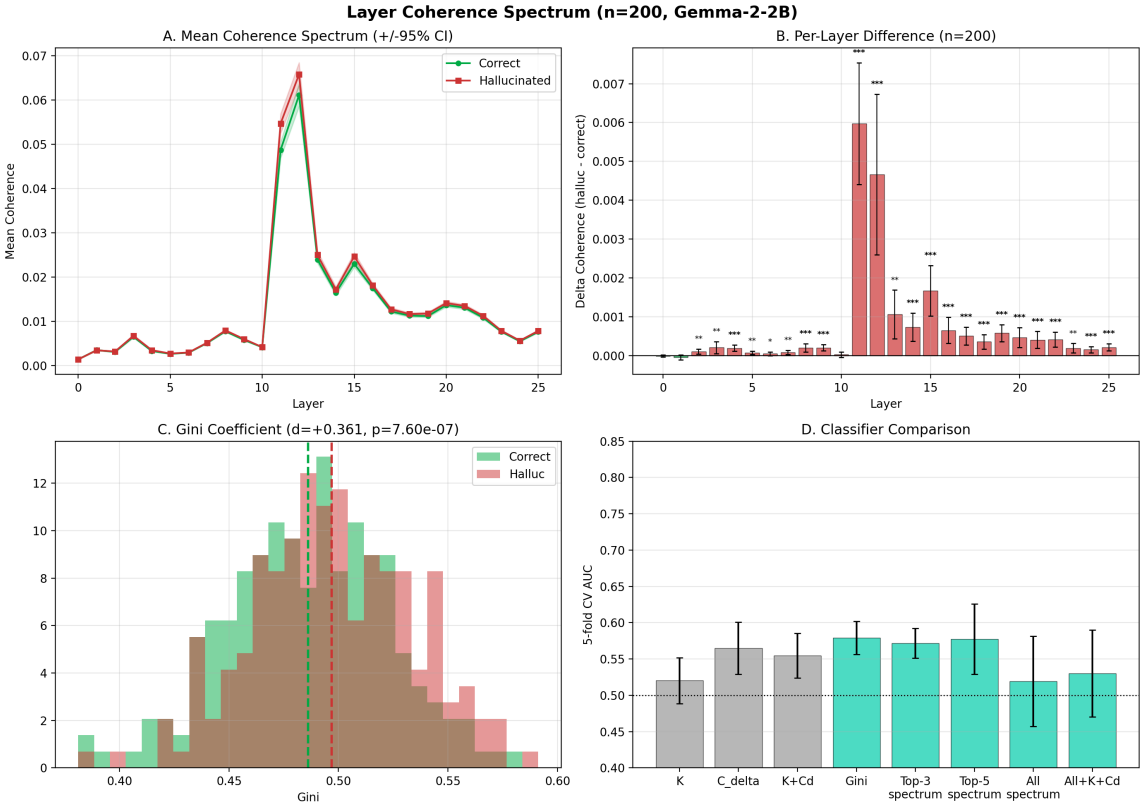


Figure 9: *Coherence spectrum analysis ($n = 200$, Gemma-2-2B only). Panel (A): mean per-layer coherence spectra with 95% confidence bands. Panel (B): per-layer difference with significance markings (16/26 layers at $p < 0.001$). Panel (C): Gini coefficient distributions. Panel (D): 5-fold CV AUC classifier comparison.*

tokens) and its cross-architecture replication on two architecture families.

C.1 S_{early} cross-dataset meta-analysis

S_{early} was evaluated on three independent datasets with different experimental designs (Table 7). The effect sizes are consistent ($d \in [0.466, 0.506]$) and Fisher combined $p = 0.0008$ (one-sided). Subsample robustness: 95.8% of 50%-subsamples show $d > 0$ across all three datasets. The paired hedge-vs-commit experiments are length-matched by construction, ruling out question-length confounds that could affect the TruthfulQA comparison.

C.2 Head-to-head vs. logit entropy

On the same 200 TruthfulQA items (Gemma-2-2B-IT), we captured both S_{early} and standard behavioural uncertainty measures per token (Table 8). S_{early} is the only feature reaching conventional significance on this sample. The correlation with logit entropy is $r = -0.17$ (nearly independent).

Dataset	Design	d	AUC
Gemma-2-2B base hedge/commit	paired, $n = 30$	+0.501	0.66
Gemma-2-2B-IT hedge/commit	paired, $n = 30$	+0.466	0.63
Gemma-2-2B-IT TruthfulQA (hall vs corr)	unpaired, $n = 94$	+0.506	0.67
Pooled meta-effect		**+0.494**	

Table 7: *S_{early} cross-dataset meta-analysis. Fisher combined $p = 0.0008$. Effect sizes consistent at $d \approx 0.5$ across three independent designs. The paired designs are length-matched.* ★ *truthfulqa_results/S_baselines_n200.json*

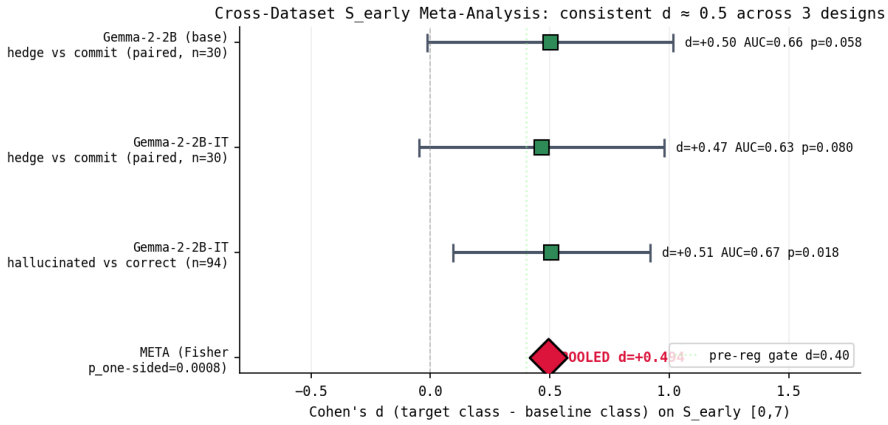


Figure 10: *Forest plot of the cross-dataset S_{early} meta-analysis. Three independent datasets, Fisher combined $p = 0.0008$.*

C.3 SAE necessity: three SAE-free probes fail

To establish that the SAE decomposition is doing irreplaceable work, we tested three SAE-free probes on four architecture families (Gemma, Llama, Qwen, Phi), each on 200 TruthfulQA items. The probes were: raw top-k IPR on the logit distribution, cross-layer cosine of raw residual streams, and a multi-metric fingerprint combining six standard uncertainty quantities. All three SAE-free probes fail at Fisher combined $p > 0.6$, mean 5-fold CV AUC = 0.566 (Table 9). The SAE-based S instrument succeeds on both tested architectures (Gemma $d = +0.51$, Llama $d = +0.72$, Fisher combined $p = 0.0006$).

C.4 Cross-architecture replication of S_{early}

S_{early} was replicated across architecture families with architecture-adaptive token windows (Table 10). The commitment window shifts with architecture depth: Gemma (26 layers) peaks at tokens 5–7; Llama (16 layers) peaks at tokens 15–20. Same direction, same AUC, different timing.

C.5 The 2D cognitive state map (Gemma-2-2B-IT)

Combining S_{early} and logit entropy into a 2D space (Figure 11) reveals four cognitive zones. The danger zone (high-S, high-entropy) is 65% hallucinated; the safe zone (low-S, low-entropy) is 22%

Feature	AUC	p	Type
S_{early}	**0.663**	**0.013**	SAE-derived
$\text{entropy}_{\text{max,early}}$	0.607	0.053	behavioural
$\text{entropy}_{\text{mean,early}}$	0.596	0.133	behavioural
$\text{logprob}_{\text{mean,early}}$	0.559	0.291	behavioural
$\text{margin}_{\text{mean,early}}$	0.477	0.624	behavioural

Table 8: *Head-to-head comparison of S_{early} against standard behavioural uncertainty measures on the same 200 TruthfulQA items (Gemma-2-2B-IT).* ★ *truthfulqa_results/baselines_headtohead.json*

Probe	Fisher p (4 archs)	Result
Top-k IPR (raw logit)	0.619	Mixed directions
Cross-layer cosine (residual)	0.948	All $d < 0$
Multi-metric fingerprint	—	Mean CV AUC = 0.566
SAE-based S (Gemma)	**0.018**	$d = +0.51$
SAE-based S (Llama)	**0.012**	$d = +0.72$

Table 9: *SAE-free probes fail; SAE-based S succeeds on both tested architectures with public SAEs available at the time of the study.* ★ *truthfulqa_results/cross_arch_meta.json*

hallucinated; $\chi^2 = 7.17$, $p = 0.007$. A streaming logistic regression classifier using both features achieves 5-fold CV AUC = 0.701 at token 7 on this model.

D Pre-registration chronology (git commits)

All commits on `master` on 2026-04-10, Eastern Time.

```
a122810  audit(atlas): bulletproof rigor pass + scope correction of v0.2.1
e32cc75  prereg(atlas): v0.3 attractor replication -- SEALED
01969cb  feat(atlas): v0.3 expansion to n=6 families -- H1 SUPPORTED
b09e54f  chore: finalization pass before Zenodo v14 / OSF upload
c396a83  release: Paper v14 + Atlas v0.3 published to Zenodo
```

Between `e32cc75` (pre-registration sealed at 14:57:52 ET) and `01969cb` (v0.3 captures + analysis committed at 16:30:28 ET), elapsed wall-clock time was 93 minutes. The analysis script `atlas/analysis/atlas_attractor_n5.py` was first committed in `01969cb` and has not been modified since. Any divergence between the committed analyzer and the sealed decision rule in the pre-registration would be visible in the git history.

E Deviations from pre-registration

The pre-registration Section 7 enumerates allowed deviations. Three substitutions were made, all under the allowed list:

Model	Layers	Window	d	AUC
Gemma-2-2B-IT	26	$[0, 7)$	$+0.506$	0.669
Llama-3.2-1B-Instruct	16	$[0, 20)$	$+0.717$	0.668
Pooled			$+\mathbf{0.587}$	

Table 10: *S_{early} cross-architecture replication with adaptive windows. Fisher combined $p = 0.0006$ on the two tested architectures with public SAEs.* ★ *truthfulqa_results/S_truthfulqa_llama_n200.json*

- **Phi-3.5-mini-instruct** was planned as an independent family. It is not in the TransformerLens registry at our version. Substituted with **Gemma-3-1B**, which is a different generation from Gemma-2-2B.
- **SmolLM2-1.7B-Instruct** was planned. It was functional but we de-prioritized it in favor of a direct scale-variant study. **Llama-3.2-1B** was substituted as a scale variant of the already-validated Llama-3.2-3B.
- **OLMo-2-1B** was planned. Not attempted this session due to time budget. **Qwen2.5-1.5B** was substituted as a scale variant of the already-validated Qwen2.5-3B.

The substitutes are scale or generation variants of the original three v0.2 families rather than fully independent families. This makes the v0.3 replication weaker than originally planned, which we state explicitly in §8. The v0.4 target is truly independent families.

F IPR physics derivation in full

We give the full IPR derivation for completeness.

Setup. Let $C_\Delta(t)$ be the coherence delta at token t for $t = 1, \ldots, T$. Fix a threshold $\tau = 0.005$. Define the event indicator set $\mathcal{E} = \{t : C_\Delta(t) > \tau\}$ and the event count $K_\tau = |\mathcal{E}|$. Define the uniform event-location distribution $\hat{e}_t = 1/K_\tau$ for $t \in \mathcal{E}$ and 0 otherwise.

Participation ratio and IPR. The participation ratio of a non-negative distribution $\hat{e}$ over n sites is $\mathrm{PR}(\hat{e}) = (\sum_t \hat{e}_t)^2 \,/\, \sum_t \hat{e}_t^2 = 1 \,/\, \sum_t \hat{e}_t^2$ when $\hat{e}$ is a probability distribution. The inverse participation ratio is $\mathrm{IPR}(\hat{e}) = \sum_t \hat{e}_t^2 = 1 \,/\, \mathrm{PR}$.

IPR of the uniform event distribution. For the uniform distribution over K_τ sites, $\hat{e}_t = 1/K_\tau$ for each active position and 0 otherwise. Then

$$\mathrm{IPR}(\hat{e}) = \sum_{t \in \mathcal{E}} \left(\frac{1}{K_\tau}\right)^2 = K_\tau \cdot \frac{1}{K_\tau^2} = \frac{1}{K_\tau}.$$

Fathom commitment intensity. Define

$$S = \max_t C_\Delta(t) \cdot \mathrm{IPR}(\hat{e}) = \frac{\max_t C_\Delta(t)}{K_\tau}.$$

This is exactly the Fathom definition in §3. The equivalence is algebraic and holds to machine precision on any trajectory with $K_\tau > 0$.

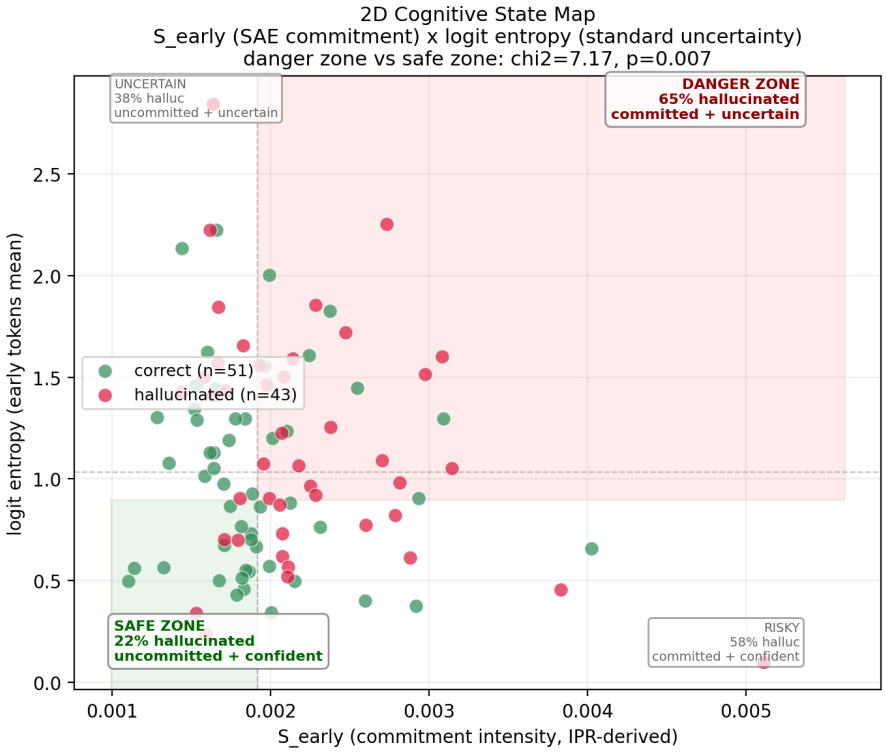


Figure 11: *The 2D cognitive state map on Gemma-2-2B-IT TruthfulQA. Each generation is a point in (S_{early}, entropy) space. Danger zone (high-S, high-entropy), 65% hallucinated; safe zone (low-S, low-entropy), 22% hallucinated.* $\chi^2 = 7.17$, $p = 0.007$.

Why the ratio form is specific. The cross-dataset meta- analysis showed that only the ratio replicates across all three datasets, while the max alone fails (pooled $d = +0.309$, inconsistent sign across datasets) and the mean alone fails (pooled $d = +0.067$, no signal). The IPR framing explains this: max captures peak magnitude but discards event sparsity, and mean captures neither. Only the ratio captures both, and that is exactly what the IPR prescribes. Metric specificity is a signature of a real physical construct, not a post-hoc-tuned detector.

Historical context. The IPR appears in Anderson [1] as the diagnostic for localization in disordered quantum systems. It has been refined in Edwards–Thouless theory, extended to random matrix eigenvector statistics, and is a standard tool in quantum many-body physics and disordered systems analysis. The Fathom commitment intensity is a specific instance of this construct applied to the coherence event distribution in transformer internals.